

The general notion of an eigenvalue

Definition

Suppose that $T : V \rightarrow V$ is a linear transformation defined on a vector space V , If $T(v) = \lambda v$ for some $v \in V$ which is not zero we say that λ is an eigenvalue for T and v is an eigenvector for λ .

① If V is fin. dim then T is T_A for some matrix A
and everything we did up to now applies.

② $V =$ all inf. diff. functions on $\mathbb{R}$.

$$D: V \rightarrow V \text{ where } D(f) = f'$$

For what λ does $D(f) = \lambda f$? or $f' = \lambda f$.

Give me an f st. $f' = \lambda f$

$$f(x) = e^{\lambda x} \text{ then } f' = \lambda e^{\lambda x} = \lambda f$$

So D has every $\lambda \in \mathbb{R}$ as an eigenvalue.

③ $V =$ the v. space of polys.

$$T: V \rightarrow V \text{ st. } T(p) = xp$$

What are the eigenvalues for T ?

We want $\underline{xp} = \underline{\lambda p}$ for some λ and $p \neq 0$.

No λ works!

Matrix associated to a linear transformation

- 1 Suppose that V is finite-dimensional with basis $B = \{b_1, \dots, b_n\}$, and $T : V \rightarrow V$ is a linear transformation.
- 2 We can compute $T(b_1), \dots, T(b_n)$ and consider their B -coordinates.
- 3 If we let A be the matrix

$$A = [T(b_1)_B \quad T(b_2)_B \quad \dots \quad T(b_n)_B] \quad \leftarrow \begin{matrix} n \times n \\ \text{matrix} \end{matrix}$$

then for any $v \in V$, $T(v)_B = Av_B$.

Ex: $T: P_2 \rightarrow P_2$, $B = \{1, x, x^2\}$.

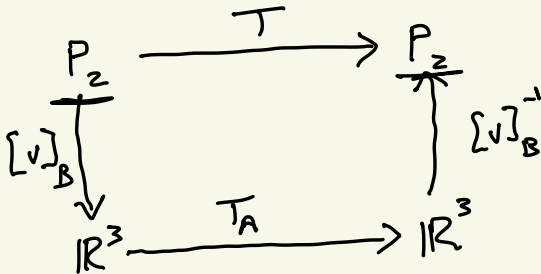
$T(p(x)) = p(x+1)$ What is the matrix A here?

$$T(1) = \underline{1}, \quad T(x) = \underline{x+1}, \quad T(x^2) = \underline{(x+1)^2} = \underline{x^2 + 2x + 1}$$

$$[1]_B = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad [x+1]_B = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \quad [x^2+2x+1]_B = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$$

Eigenvalue(s) for T
is 1 with mult. 3.



Compute $T(x^2 - 3x + 2)$

$$= (x+1)^2 - 3(x+1) + 2$$

$$= x^2 - x$$

$$[x^2 - 3x + 2]_B = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} = [Av]_B$$

$\uparrow$
 v

so $T(x^2 - 3x + 2) = 0 \cdot 1 + (-1)x + 1 \cdot x^2 = x^2 - x.$

$$V = \text{span}(\sin(x), \cos(x)) \in 2\text{-dim.}$$

$$T: V \rightarrow V, \quad T(f) = f'.$$

$$T(\sin(x)) = \cos(x), \quad T(\cos(x)) = -\sin(x)$$

$$\begin{bmatrix} \cos(x) \end{bmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \begin{bmatrix} -\sin(x) \end{bmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

$$\text{so } A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \begin{matrix} (2\sin(x) + \cos(x))' \\ \downarrow \\ \begin{pmatrix} 2 \\ 1 \end{pmatrix} \end{matrix}$$

$$A \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix} \quad \begin{pmatrix} -1 \\ 2 \end{pmatrix} \text{ as a function is } -\sin(x) + 2\cos(x).$$

Change of basis matrix

- 1 Suppose we have a basis $B = \{b_1, \dots, b_n\}$ for $\mathbb{R}^n$.
- 2 Consider the identity map $id : \mathbb{R}^n \rightarrow \mathbb{R}^n$ where we wish to consider the domain written in standard coordinates and the codomain written in B coordinates.
- 3 The matrix for this transformation looks like

$$P = \begin{bmatrix} [e_1]_B & [e_2]_B & \dots & [e_n]_B \end{bmatrix}$$

The dot product

If $u = \begin{bmatrix} u_1 \\ \dots \\ u_n \end{bmatrix}$ and $v = \begin{bmatrix} v_1 \\ \dots \\ v_n \end{bmatrix}$ then

$$(u_1 \dots u_n) \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$$

$$u \cdot v = u_1 v_1 + \dots + u_n v_n.$$

We could also we write $u \cdot v = u^T v$.

Fact

Suppose that u, v and w are in $\mathbb{R}^n$. Then

- 1 $u \cdot v = v \cdot u,$
- 2 $u \cdot (v + w) = u \cdot v + u \cdot w,$
- 3 $(cu) \cdot v = c(u \cdot v)$ for any $c \in \mathbb{R}$, and
- 4 $u \cdot u \geq 0$ and $u \cdot u = 0$ iff $u = 0$.

$$\underline{u \cdot u = u_1^2 + u_2^2 + \dots + u_n^2 \geq 0}$$

Length and distance in $\mathbb{R}^n$

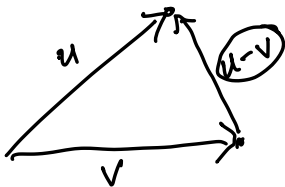
For $u \in \mathbb{R}^n$, the length of $u = (u_1, \dots, u_n)^T$ is

$$\|u\| = \sqrt{u_1^2 + \dots + u_n^2}. \quad \leftarrow \sqrt{u \cdot u}$$

The distance between two vectors u and v , $d(u, v)$ is $\|u - v\|$.

The angle θ between non-zero vectors u and v in $\mathbb{R}^n$ satisfies

$$\cos(\theta) = \frac{u \cdot v}{\|u\| \|v\|}.$$



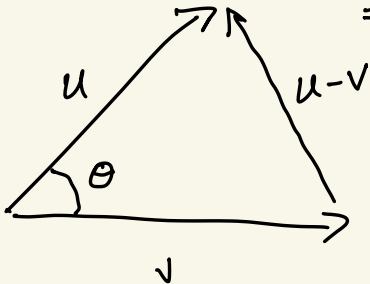
$$\|u - v\| = d(u, v).$$

$$\underline{\|u-v\|^2 = \frac{\|u\|^2 + \|v\|^2 - 2\|u\|\|v\|\cos\theta}{}}$$

$$\|u-v\|^2 = (u-v) \cdot (u-v)$$

$$= u \cdot u - u \cdot v - v \cdot u + v \cdot v$$

$$= \underline{\|u\|^2 - 2u \cdot v + \|v\|^2}$$



$$\stackrel{\text{def}}{=} u \cdot v = \|u\| \|v\| \cos\theta$$

$$\text{or } \cos\theta = \frac{u \cdot v}{\|u\| \|v\|}$$

Orthogonality

In $\mathbb{R}^n$ we say that u and v are orthogonal if the angle between them is $\frac{\pi}{2}$ or equivalently if $u \cdot v = 0$.

In $\mathbb{R}^n$ we have the Pythagorean Theorem: If u and v are orthogonal then

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2.$$

Orthogonal complement

Suppose that W is a subspace of $\mathbb{R}^n$, the orthogonal complement of W , written $W^\perp$ is defined by

$$W^\perp = \{u \in \mathbb{R}^n : u \cdot w = 0 \text{ for all } w \in W\}.$$

For any subspace W of $\mathbb{R}^n$, $W^\perp$ is a subspace of $\mathbb{R}^n$ as well.
 $W \cap W^\perp = \{0\}.$

Orthogonal sets and basis

Suppose that $v_1, \dots, v_k$ are non-zero vectors in $\mathbb{R}^n$. We say $v_1, \dots, v_k$

- is an orthogonal set if v_i and v_j are orthogonal for all $i \neq j$.
- is an orthonormal set if it is orthogonal and each vector has length 1.
- is an orthogonal (orthonormal) basis if it is a basis for $\mathbb{R}^n$ as well as being orthogonal (orthonormal).

Fact

Any orthogonal set in $\mathbb{R}^n$ is linearly independent.